

Answer all questions from each group

Group A

5x1=5

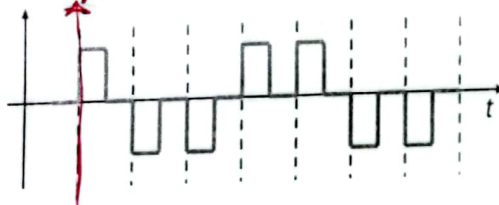
Q1 Answer any **FIVE**

- Write two instances of how networks are a part of your daily life today.
- A network contains four computers. If there are only four lengths of cable in this network, which topology is used?
- Write the name of two protocols of the Application layer.
- What is the binary representation of $x^8 + x^3$?
- What is the result of shifting 111110 four bits to the right?
- Write one difference between FHSS and DSSS.
- What is the propagation method for Radio wave?
- What are the values of SNR and SNRDB for a noiseless channel?

4x2.5=10

Q2 Answer any **FOUR**

- Draw the constellation diagram for the following:
 - 8-QAM with two different amplitudes and four different phases.
 - 16-QAM with two different amplitudes and eight different phases.
- A non-periodic composite signal contains frequencies from 10 to 30 KHz. The peak amplitude is 10 V for the lowest and the highest signals and 40 V for the 20-KHz signal. Assuming that the amplitudes change gradually from the minimum to the maximum, draw the frequency spectrum.
- Find the ~~8~~ bit data stream for the RZ scheme depicted in the following figure:



- A network with a bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?
- What is the minimum Hamming distance? If we want to be able to detect two-bit errors, what should be the minimum Hamming distance?
- We need a three-stage space-division switch with $N = 100$. We use 10 crossbars at the first and third stages and 4 crossbars at the middle stage.
 - Draw the configuration diagram.
 - Calculate the total number of crosspoints.

2x5=10

Q3 Answer any **TWO**

- Four 1Mbps input lines are multiplexed in a synchronous TDM. The unit of data is 1bit. Find
 - The input bit duration
 - The output bit duration
 - The output bit rate
 - The output frame rate
- What is the result of scrambling the sequence 11100000000000 using B8ZS? Assume that the last non-zero signal level has been positive.
 - Draw the graph of the MLT-3 scheme for 00011000, assuming that the last signal level has been negative.
- Suppose you want to send a data of five 8-bit numbers: 16, 19, 25, 0, 9. Calculate wrapped sum and checksum. If there is no error occurred in transmission, show that the data have not been corrupted on the receiver side.

Group B

Q4 Answer any FIVE

5x1=5

- What is the difference between STP and UTP?
- What is a Multicast Address? Give an example.
- What is Pulse Stuffing?
- Which of the four digital-to-analog conversion techniques (ASK, FSK, PSK, or QAM) is the most susceptible to noise? Defend your answer.
- What is Nyquist Bit Rate?
- What is baseline wandering?
- What is the hamming distance for each of the following codewords?
 - d(10000, 11010)
 - d(11111, 10000)
- What is the polynomial representation of 10110011?

Q5 Answer any FOUR

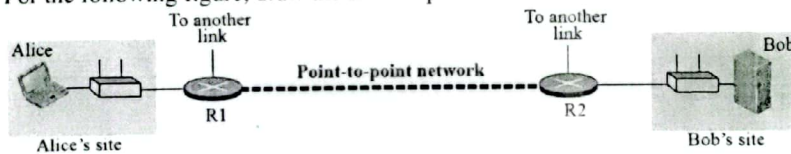
4x2.5=10

- Draw the graph of Differential Manchester for the 00110001 assuming that the last signal level has been positive.
- A TV channel has a bandwidth of 6 MHz. If we send a digital signal using one channel, what are the data rates if we use one harmonic, three harmonics, and five harmonics?
- An analog signal has a bit rate of 8000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements do we need?
- We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What are the carrier frequency and the bit rate if we modulated our data by using ASK with $d=1$?
- Why does a router have more than one interface? How many IP addresses and how many link-layer addresses should a router have when it is connected to three links?
- Assuming even parity, find the parity bit for 11001100.

Q6 Answer any TWO

2x5=10

- Consider a sample signal with values in the order: -6.1, 7.5, 17.5, 19.7, 11.0, -5.5, -11.3, -9.4, and -6.0. The signal's amplitudes are between -20V to +20V. If there are 8 levels, then find out
 - Normalized PAM values
 - Normalized quantized values
- For the following figure, draw the flow of packets at Router R1.



- Draw the flow diagram of the Stop and Wait Protocol with Sequence Number and Acknowledgment numbers using the following scenario:
 - Frame 0 is sent, but lost.
 - Frame 0 is resent and acknowledged.
 - Frame 1 is sent and acknowledged, but the acknowledgment is lost.
 - Frame 1 is resent, but it is timed-out.
 - Frame 1 is resent and acknowledged.